

ZOMATO

SALES & BUSINESS ANALYTICS

SQL-Based Business Analysis and Interactive Power BI Dashboard

Project Report

Tools Used: MySQL | SQL | Power BI | DAX | Excel

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1. Project Overview

This project analyzes Zomato order and delivery data to identify patterns in sales, customer behavior, restaurant performance, delivery operations, and city-level business performance.

The analysis was performed using MySQL for data exploration and business analysis, followed by Power BI for interactive visualization and dashboard development.

The project converts raw transactional data into actionable business insights that can support decisions related to revenue growth, delivery efficiency, restaurant performance, customer experience, and operational improvement.

2. Business Problem Statement

Food-delivery platforms generate large volumes of transactional data related to orders, restaurants, customers, payments, delivery operations, and locations. However, raw data alone does not clearly reveal:

- Which cities generate the most business?
- Which restaurants contribute the highest revenue?
- How does delivery performance vary across cities?
- What is the relationship between delivery time and customer ratings?
- Which restaurants have high order volumes but lower ratings?
- How does delivery distance relate to order value?
- What patterns exist in customer tipping and payment behavior?

The objective of this project is to analyze these areas and present the findings through an interactive Power BI dashboard that enables business users to explore performance and identify potential areas for improvement.

3. Project Objectives

1. Analyze overall sales and order performance.
2. Identify high-performing cities and restaurants.
3. Evaluate delivery time and distance.
4. Analyze customer ratings and experience.
5. Examine payment and order-status patterns.
6. Analyze order value and tipping behavior.
7. Identify operational and business-performance patterns.

8. Develop an interactive Power BI dashboard.
9. Convert analytical findings into actionable business insights.

4. Tools & Technologies

Tool	Purpose
Excel	Initial dataset
MySQL	Data storage and SQL analysis
SQL	Business analysis and KPI calculations
Power BI	Interactive dashboard
DAX	Measures and calculated columns
Power BI Visuals	Data visualization and analysis

5. Dataset Description & Data Preparation

The dataset represents Zomato order-level transactional records, with each row corresponding to a single customer order. The core fields used throughout the SQL analysis and Power BI dashboard include:

- `order_amount_inr` — the monetary value of each order (INR)
- `delivery_time_mins` — time taken from order placement to delivery
- `delivery_distance_km` — distance between restaurant and customer
- `rating` — customer rating given for the order
- `tip_amount_inr` — tip amount given by the customer
- `order_status` — order outcome (e.g., Delivered, Cancelled)
- city, restaurant, and payment method fields used for segmentation

Data Preparation Steps

1. Imported the raw dataset from Excel into MySQL for structured storage.
2. Standardized data types for numeric fields (order amount, delivery time, distance, ratings, tips).
3. Checked for and handled missing, blank, or inconsistent values in key columns.
4. Removed duplicate order records to ensure accuracy of aggregated KPIs.
5. Created categorical groupings (delivery-time buckets, distance buckets, tip buckets) to support segmentation analysis.
6. Validated referential consistency between orders, restaurants, and city fields prior to analysis.

6. SQL-Based Business Analysis

Structured Query Language (SQL) in MySQL was used to explore the dataset and calculate the core business metrics that later powered the Power BI dashboard. The SQL analysis focused on the following areas:

- Aggregating total revenue, order counts, and average order value at the overall, city, and restaurant level.
- Segmenting orders into delivery-time categories (0–20, 21–30, 31–45, 46–60, 60+ minutes) and distance categories (0–2 km, 2–5 km, 5–10 km, 10+ km).
- Calculating the correlation between delivery distance and delivery time.
- Identifying high-volume restaurants with average ratings below key thresholds.
- Categorizing customer tipping behavior into defined tip ranges.
- Summarizing orders by payment method and order status (e.g., Delivered vs. Cancelled).

Representative SQL Queries

A selection of the core queries used during the analysis is shown below.

1. City-wise Revenue

```
SELECT city,
       COUNT(*) AS total_orders,
       SUM(order_amount_inr) AS total_revenue,
       ROUND(AVG(order_amount_inr),2) AS avg_order_value
FROM zomato_orders
GROUP BY city
ORDER BY total_revenue DESC;
```

2. Top 10 Restaurants by Revenue

```
SELECT restaurant_name,
       SUM(order_amount_inr) AS total_revenue,
       COUNT(*) AS total_orders,
       ROUND(AVG(rating),2) AS avg_rating
FROM zomato_orders
GROUP BY restaurant_name
ORDER BY total_revenue DESC
LIMIT 10;
```

3. Delivery Performance by Time Category

```
SELECT
  CASE
    WHEN delivery_time_mins <= 20 THEN '0-20 mins'
    WHEN delivery_time_mins <= 30 THEN '21-30 mins'
    WHEN delivery_time_mins <= 45 THEN '31-45 mins'
    WHEN delivery_time_mins <= 60 THEN '46-60 mins'
    ELSE '60+ mins'
  END AS delivery_time_category,
  COUNT(*) AS total_orders,
  ROUND(AVG(rating), 2) AS avg_rating
FROM zomato_orders
GROUP BY delivery_time_category
ORDER BY MIN(delivery_time_mins);
```

4. Payment Method Analysis

```
SELECT payment_method,
       COUNT(*) AS total_orders,
       SUM(order_amount_inr) AS total_revenue,
       ROUND(AVG(order_amount_inr), 2) AS avg_order_value,
       ROUND(100 * COUNT(*) /
         (SELECT COUNT(*) FROM zomato_orders), 2) AS pct_of_orders
FROM zomato_orders
GROUP BY payment_method
ORDER BY total_revenue DESC;
```

5. Correlation: Delivery Distance vs Delivery Time

```
SELECT
  (COUNT(*) * SUM(delivery_distance_km * delivery_time_mins)
   - SUM(delivery_distance_km) * SUM(delivery_time_mins))
  /
  (SQRT(COUNT(*) * SUM(POW(delivery_distance_km,2))
```

```
- POW(SUM(delivery_distance_km),2))
* SQRT(COUNT(*) * SUM(POW(delivery_time_mins,2))
- POW(SUM(delivery_time_mins),2)))
AS correlation_distance_time
FROM zomato_orders;
```

These queries formed the analytical foundation that was then connected to Power BI, where DAX measures replicated and extended the same logic for interactive, filterable reporting.

7. Power BI Dashboard Development

After completing the SQL analysis, the processed data was connected to Microsoft Power BI to create an interactive dashboard. DAX measures and calculated columns were created to support KPI cards, filtering, segmentation, and business-focused visualizations.

The dashboard was divided into three pages, with each page focusing on a different aspect of the business.

7.1 DAX Measures Created

```
Total Revenue =  
SUM(zomato_orders[order_amount_inr])  
  
Total Orders =  
COUNTROWS(zomato_orders)  
  
Average Order Value =  
AVERAGE(zomato_orders[order_amount_inr])  
  
Average Delivery Time =  
AVERAGE(zomato_orders[delivery_time_mins])  
  
Average Customer Rating =  
AVERAGE(zomato_orders[rating])  
  
Total Tips =  
SUM(zomato_orders[tip_amount_inr])  
  
Average Delivery Distance =  
AVERAGE(zomato_orders[delivery_distance_km])  
  
Cancelled Orders =  
CALCULATE(  
    COUNTROWS(zomato_orders),  
    zomato_orders[order_status] = "Cancelled"  
)
```

These measures provided dynamic calculations that automatically respond to dashboard filters.

7.2 Page 1 — Executive Overview

The first page was designed to provide a high-level view of overall business performance.

KPI Cards

- Total Revenue
- Total Orders
- Average Order Value
- Average Delivery Time
- Average Customer Rating

Visualizations

- Monthly Revenue Trend
- Revenue by City
- Top 10 Restaurants by Revenue
- Total Orders by Order Status
- Order Value by Status
- Orders by Payment Method

Interactive Filters

- City
- Order Date
- Order Status

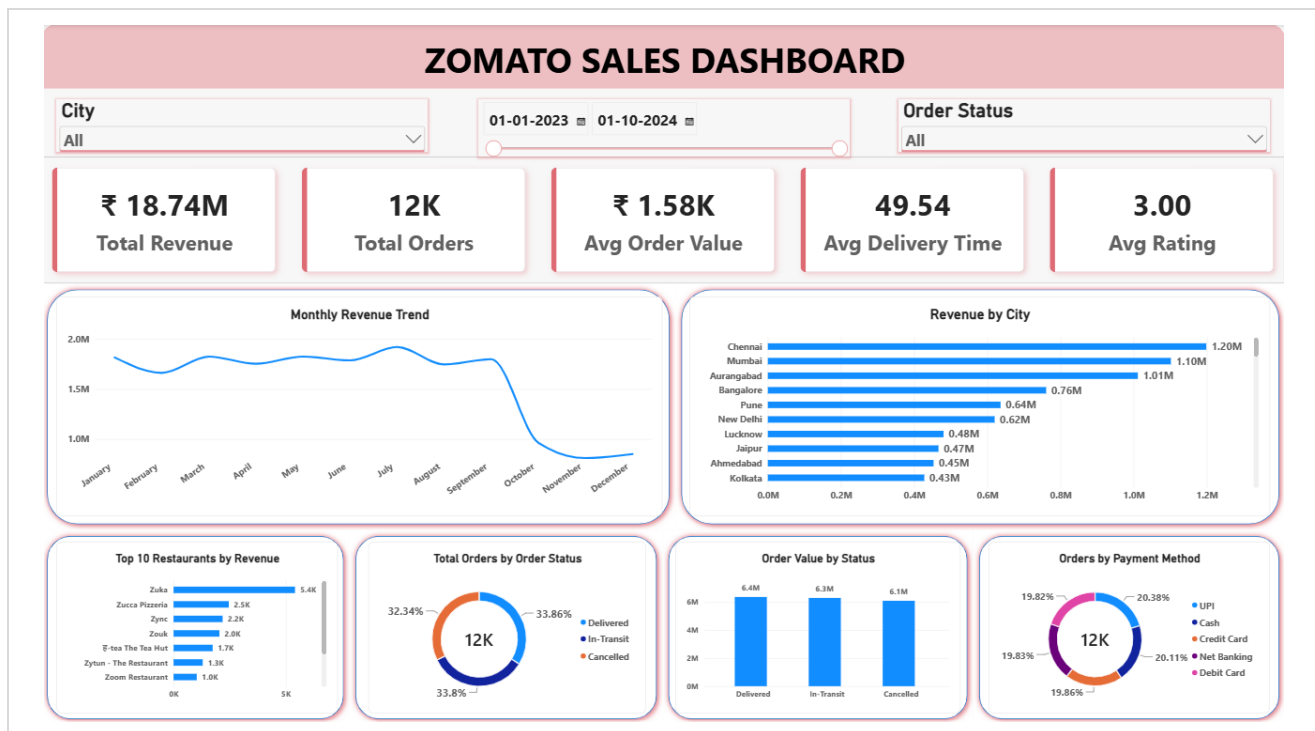


Figure 1 — Power BI Dashboard, Page 1: Executive Overview

7.3 Page 2 — Operations & Customer Experience

The second page focuses on delivery operations and customer experience.

KPI Cards

- Average Delivery Time
- Average Delivery Distance
- Average Customer Rating
- Total Orders

Visualizations

- Orders by Delivery Time Category
- Delivery Time vs Customer Rating
- Orders by Day of Week
- Delivery Distance vs Delivery Time
- Average Delivery Time by City

Filters

- City
- Order Status
- Order Date (Between range slicer for custom periods)

This page helps analyze delivery efficiency, customer experience, and operational differences between cities.

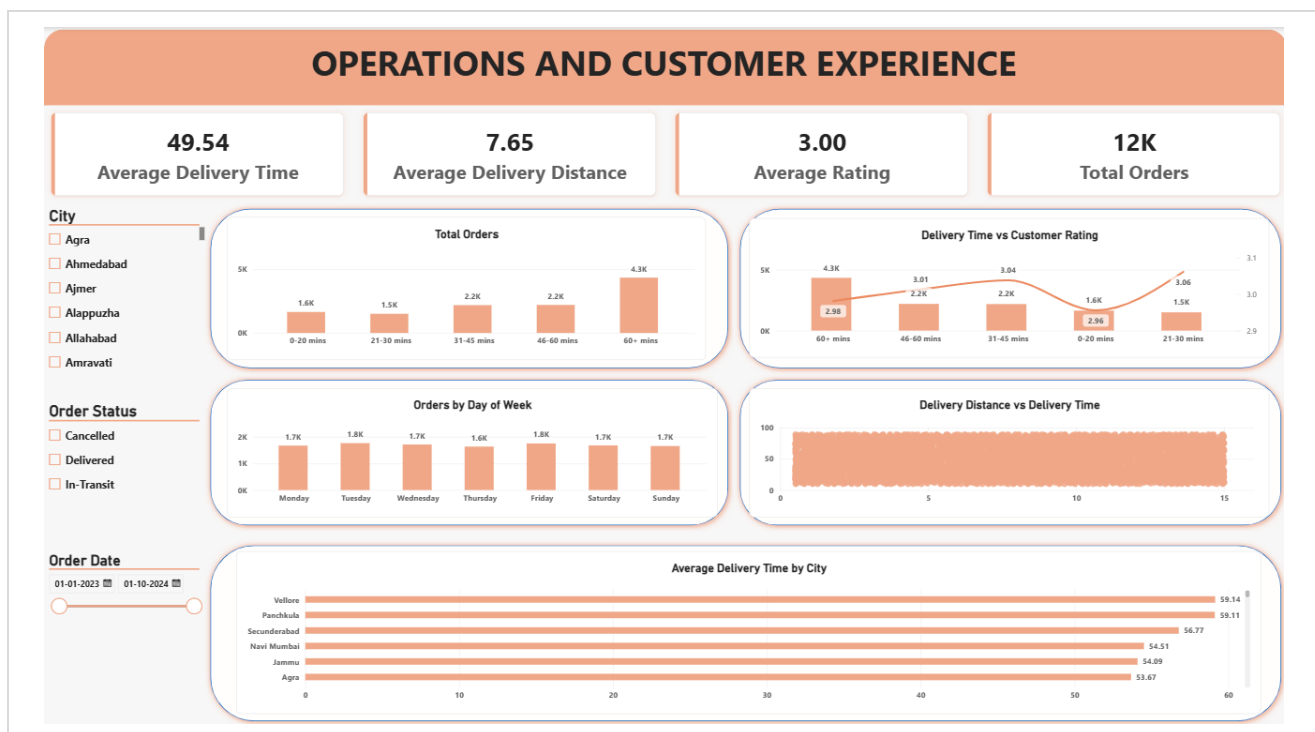


Figure 2 — Power BI Dashboard, Page 2: Operations & Customer Experience

7.4 Page 3 — Actionable Business Insights

The third page was designed to move from descriptive analysis toward business-oriented insights.

Visualizations

- City Revenue vs Order Volume
- High-Revenue Cities with Slow Delivery
- Top 10 Restaurants by Revenue

- High-Volume Restaurants with Low Ratings
- Average Order Value by Delivery Distance
- Orders by Tip Category

Business Insight Section

Delivery Performance: High-revenue cities with elevated delivery times can be investigated for operational improvements.

Restaurant Quality: High-volume restaurants with ratings below 3.0 can be investigated for customer-experience issues.

Order Value: Order value can be compared across delivery-distance categories to understand customer purchasing patterns.

Customer Tipping: The distribution of tip categories can be analyzed to understand factors associated with higher customer tipping.

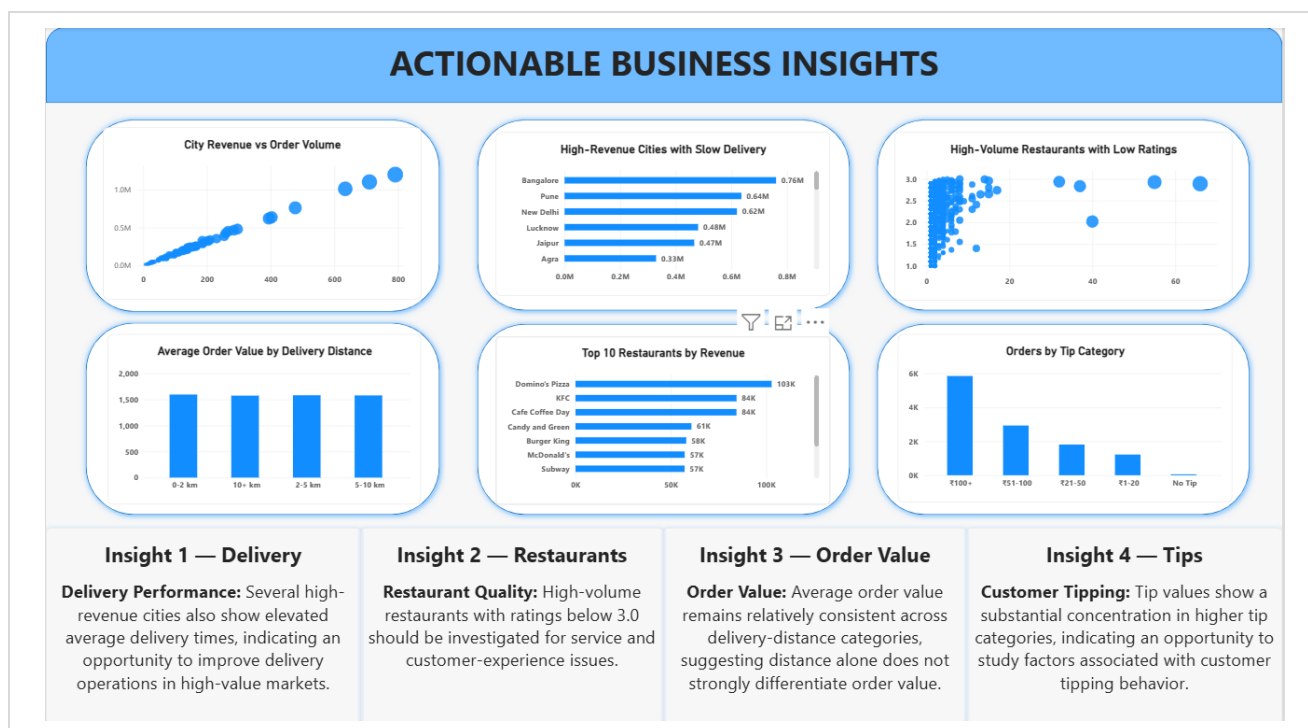


Figure 3 — Power BI Dashboard, Page 3: Actionable Business Insights

7.5 Dashboard Interactivity

The dashboard uses Power BI's interactive filtering capabilities to allow users to:

- Filter by city
- Filter by order status
- Select different date ranges
- Compare restaurant performance
- Explore delivery performance

- Examine customer experience indicators

This transforms the dashboard from a static report into an interactive business-analysis tool.

7.6 Dashboard Development Workflow

```
MySQL Data -> Power BI Connection -> DAX Measures -> Calculated Columns ->
Visualizations -> Slicers -> Business Insights
```

This structure allows raw transactional data to be converted into an interactive analytical dashboard.

8. Key Findings & Business Insights

The analysis of Zomato order data produced several findings across sales performance, delivery operations, restaurant performance, customer experience, and customer behavior.

8.1 Sales & Revenue Performance

- Revenue and order volume were analyzed across different cities to identify variations in market performance.
- City-level analysis helps identify markets that contribute significantly to overall business activity.
- Restaurant-level revenue analysis highlighted the contribution of individual restaurant partners.
- Average Order Value was used to understand the typical transaction size and compare purchasing patterns.

Business implication: High-performing cities and restaurants can be analyzed further to understand the factors contributing to their performance and identify practices that could potentially be replicated in other markets.

8.2 Delivery Performance

Delivery time was analyzed using multiple time categories:

- 0–20 minutes
- 21–30 minutes
- 31–45 minutes
- 46–60 minutes
- 60+ minutes

This segmentation made it possible to compare order volumes and customer ratings across different delivery-time ranges. The analysis also compared delivery performance across cities.

Business implication: Cities with relatively high delivery times can be examined for operational factors such as delivery-agent availability, restaurant preparation time, traffic conditions, and order allocation.

8.3 Delivery Distance & Delivery Time

A correlation analysis was conducted between delivery distance and delivery time. The calculated correlation was approximately:

Correlation (Delivery Distance, Delivery Time) ≈ -0.005

This indicates that the dataset shows almost no linear correlation between these two variables.

Business implication: Distance alone does not appear to explain delivery-time variation in this dataset. Other operational factors may therefore be more important when investigating delivery delays.

Possible areas for further investigation include:

- Restaurant preparation time
- Delivery-agent availability
- Traffic conditions
- Order batching
- Peak-hour demand

8.4 Customer Experience

Customer ratings were analyzed alongside delivery-time categories and restaurant performance. The dashboard also identifies restaurants with relatively high order volumes and lower average customer ratings.

Business implication: High-volume restaurants with lower ratings represent potential areas for investigation. Management could examine customer feedback, food quality, service consistency, and delivery-related complaints before deciding on corrective actions.

8.5 Restaurant Performance

Restaurant performance was evaluated using Total Orders, Total Revenue, Average Order Value, and Customer Rating. The Top 10 Restaurants by Revenue visualization highlights restaurants contributing significantly to sales, while a separate analysis identifies high-volume restaurants with lower customer ratings.

Business implication: Restaurant partners can be segmented based on both commercial performance and customer experience, rather than evaluating performance using revenue alone.

8.6 Delivery Distance & Order Value

Orders were divided into four distance categories, and average order value was compared across these categories:

Distance	Category
Less than 2 km	0–2 km
2–5 km	2–5 km
5–10 km	5–10 km
10 km or more	10+ km

Business implication: This analysis can help understand whether customer spending patterns differ based on delivery distance and can support further analysis of delivery economics and customer purchasing behavior.

8.7 Customer Tipping Behavior

Tip amounts were categorized into: No Tip, ₹1–20, ₹21–50, ₹51–100, and ₹100+. The dashboard shows the distribution of orders across these categories.

Business implication: The distribution can be used to investigate factors associated with higher tipping behavior, such as order value, delivery experience, distance, or customer location.

8.8 Payment & Order Status

Payment methods were compared using order volume, revenue, average order value, and percentage of total orders. Order status was also analyzed to understand the distribution of completed and cancelled orders.

Business implication: Understanding payment preferences can support payment-channel optimization, while monitoring order-status patterns can help identify areas where cancellations or unsuccessful orders may require further investigation.

8.9 Overall Business Takeaways

The analysis demonstrates that Zomato's performance should be evaluated through multiple dimensions rather than revenue alone. The most useful analytical dimensions identified in the project are:

Revenue + Order Volume + Delivery Performance + Customer Rating + Restaurant Performance + Customer Behavior

Combining these dimensions allows management to identify areas requiring further investigation and supports more data-driven operational decision-making.

9. Business Recommendations

Based on the SQL analysis and Power BI dashboard, the following recommendations can be considered to improve business performance and customer experience.

9.1 Improve Delivery Operations

Cities showing relatively high revenue along with higher delivery times should be examined in greater detail.

- Analyze delivery-agent availability during peak periods.
- Review restaurant preparation times.
- Examine order allocation and batching practices.
- Investigate location-specific operational constraints.

The objective should be to reduce unnecessary delivery delays without compromising order volume.

9.2 Monitor High-Volume Restaurants with Low Ratings

Restaurants generating substantial order volumes but receiving lower customer ratings should receive additional attention.

- Analyze customer feedback and complaints.
- Monitor food quality and service consistency.
- Review delivery-related complaints separately from restaurant-related complaints.
- Work with restaurant partners on corrective measures where appropriate.

9.3 Focus on High-Value Markets

Cities contributing substantial revenue represent important markets for business analysis.

- Monitor revenue and order trends regularly.
- Study factors behind strong performance.
- Evaluate whether successful operational practices can be applied to other markets.
- Track delivery performance alongside revenue instead of considering revenue alone.

9.4 Use Multi-Dimensional Restaurant Performance Analysis

Restaurant performance should not be evaluated only using revenue. A broader performance framework can consider:

Revenue + Orders + Average Order Value + Customer Rating

This allows restaurants to be identified according to both commercial contribution and customer experience.

9.5 Investigate Factors Behind Delivery Delays

The correlation analysis showed approximately -0.005 correlation between delivery distance and delivery time, indicating almost no linear relationship in this dataset. Instead of assuming that distance is the primary cause of delays, future analysis should examine:

- Restaurant preparation time
- Peak-hour demand
- Delivery-agent availability
- Traffic conditions
- Order batching
- Order allocation

9.6 Analyze Customer Tipping Behavior

The dashboard provides a breakdown of orders by tip category. Further analysis can combine tipping data with:

- Order value
- Delivery distance
- Delivery time
- Customer location
- Restaurant category

This can help identify patterns associated with higher customer tipping.

9.7 Strengthen Data-Driven Decision Making

The Power BI dashboard should be used as a recurring analytical tool rather than only as a one-time report. Business teams can periodically monitor:

- Revenue
- Order volume
- Delivery performance
- Customer ratings
- Restaurant performance
- Cancellation patterns
- Payment behavior

This would allow changes in business performance to be identified more quickly.

9.8 Overall Recommendation

The analysis suggests that business decisions should be based on a combination of financial, operational, and customer-experience metrics. A balanced monitoring framework can therefore be represented as:

Sales Performance -> Operational Efficiency -> Customer Experience -> Restaurant Performance -> Business Action

This approach can help convert transactional data into structured business insights and support continuous performance monitoring.

10. Conclusion

This project demonstrates an end-to-end data analytics workflow — from raw transactional data in Excel, through structured business analysis in MySQL using SQL, to an interactive, decision-support dashboard built in Power BI with DAX. The resulting three-page dashboard enables stakeholders to move beyond simple revenue reporting and evaluate the business across sales, delivery operations, restaurant quality, and customer behavior simultaneously.

The project highlights practical, portfolio-ready skills in SQL-based business analysis, KPI design, data modeling, DAX measure creation, and dashboard storytelling — skills directly applicable to Business Analyst, Data Analyst, and BI Developer roles.